

Healthcare Data Analysis Project

1. Project Overview

This project analyzes healthcare data using transactional records across various departments, patient demographics, and hospital billing. The goal is to uncover insights into patient flow, treatment costs, and department utilization to guide strategic hospital administration decisions.

2. Dataset Summary

- **Rows:** 20,000 (Admissions, Billing, Patients)
- **Columns:** Approx. 30 across 5 relational tables
- **Key Features:**
 - Patient demographics (Age, Gender, Blood Type)
 - Clinical details (Diagnosis, Department, Length of Stay, Doctor assigned)
 - Financial metrics (Total Cost, Insurance Covered, Payment Status)

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python using Pandas. Below is a snapshot of the initial exploratory data statistics and dataset shapes.

▼ Reading Excel File

Checking Shape of each Table

```
[51]: p_df = pd.read_excel("Healthcare_Analysis_Dataset.xlsx", sheet_name="Patients")
      p_df.shape

[51]: (20000, 6)

[52]: h_df = pd.read_excel("Healthcare_Analysis_Dataset.xlsx", sheet_name="Hospitals")
      h_df.shape

[52]: (10, 4)

[53]: a_df = pd.read_excel("Healthcare_Analysis_Dataset.xlsx", sheet_name="Admissions")
      a_df.shape

[53]: (20000, 8)

[54]: d_df = pd.read_excel("Healthcare_Analysis_Dataset.xlsx", sheet_name="Doctors")
      d_df.shape

[54]: (350, 5)

[55]: b_df = pd.read_excel("Healthcare_Analysis_Dataset.xlsx", sheet_name="Billing")
      b_df.shape

[55]: (20000, 9)
```

```
[75]: p_df.head(10)
```

```
[75]:
```

	patientid	firstname	lastname	age	gender	bloodtype	full_name	age_group
0	966-92-7726	Tarun	Kapoor	73	Male	O-	Tarun Kapoor	65+
1	784-20-8697	Swati	Patel	92	Female	B+	Swati Patel	65+
2	837-61-8267	Kabir	Jain	22	Male	AB+	Kabir Jain	19-35
3	882-61-6995	Vikram	Gupta	93	Male	A+	Vikram Gupta	65+
4	485-45-7437	Ishaan	Chaudhary	63	Male	AB-	Ishaan Chaudhary	51-65
5	277-14-4227	Isha	Das	35	Male	O+	Isha Das	19-35
6	904-13-8595	Rishabh	Bhat	55	Male	B+	Rishabh Bhat	51-65
7	797-77-5195	Manish	Das	20	Male	A-	Manish Das	19-35
8	982-44-9940	Vihaan	Pandey	5	Male	O-	Vihaan Pandey	0-18
9	854-53-5833	Nandini	Reddy	39	Female	B+	Nandini Reddy	36-50

Missing data was imputed using median values for continuous variables and random sampling for categorical variables. Age groups and stay categories were feature-engineered, after which the clean datasets were exported to CSV and loaded into a MySQL database.

Connecting This Data to MySQL

```
: from sqlalchemy import create_engine

username = "root"
password = "1234"
host = "localhost"
port = "3306"
database = "healthcare"

engine = create_engine(f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}")

|
p_df.to_sql("patients", engine, if_exists="replace", index=False)

a_df.to_sql("admissions", engine, if_exists="replace", index=False) # Use cleaned DataFrame

d_df.to_sql("doctors", engine, if_exists="replace", index=False)

b_df.to_sql("billing", engine, if_exists="replace", index=False)

h_df.to_sql("hospitals", engine, if_exists="replace", index=False)
```

4. Data Analysis using SQL (Business Transactions)

Structured analysis was performed in MySQL to answer key business questions, spanning revenue mapping, readmission tracking, and physician performance.

Total Gross Revenue by Hospital Region:

```
-- Q2) Calculate the total gross revenue (TotalCost) generated by each  
-- hospital region (North, South, East, etc.).
```

```
• with hospital_revenue as  
  (select a.hospitalid, sum(b.totalcost) as total_revenue  
   from billing b  
   join admissions a on b.admissionid = a.admissionid  
   group by a.hospitalid)  
  select h.region, round(sum(hr.total_revenue),2) as regional_gross_revenue  
   from hospital_revenue hr  
   join hospitals h on hr.hospitalid = h.hospitalid  
   group by h.region;
```

	region	regional_gross_revenue
▶	South	891833504.99
	West	443820859.49
	Central	330716238.07
	Northeast	266013732.09
	North	468761346.17
	East	198388182.35

Month-Over-Month Revenue Growth %:

```
-- Q9) Calculate the month-over-month percentage change in total cleared revenue
-- for the entire hospital network.
```

```
• with monthly_revenue as
  (select year(a.admissiondate) as rev_year,
    month(a.admissiondate) as rev_month, sum(b.totalcost) as current_revenue
  from billing b
  join admissions a on b.admissionid = a.admissionid
  where b.paymentstatus = 'cleared'
  group by year(a.admissiondate),month(a.admissiondate)),
  revenue_with_lag as
  (select rev_year, rev_month, current_revenue,
    lag(current_revenue) over (order by rev_year, rev_month) as prev_revenue
  from monthly_revenue)
  select rev_year, rev_month, current_revenue, prev_revenue,
    round((((current_revenue - prev_revenue) / prev_revenue) * 100, 2) as mom_percent_change
  from revenue_with_lag;
```

	rev_year	rev_month	current_revenue	prev_revenue	mom_percent_change
▶	2022	1	36729892.35500001	NULL	NULL
	2022	2	33485946.944999978	36729892.35500001	-8.83
	2022	3	42583140.15499999	33485946.944999978	27.17
	2022	4	42993679.940000005	42583140.15499999	0.96
	2022	5	40864277.15000002	42993679.940000005	-4.95
	2022	6	39176684.78	40864277.15000002	-4.13
	2022	7	43147659.29999994	39176684.78	10.14
	2022	8	40706737.83000001	43147659.29999994	-5.66
	2022	9	37338227.045	40706737.83000001	-8.28
	2022	10	37089867.165	37338227.045	-0.67
	2022	11	35740254.48999998	37089867.165	-3.64
	2022	12	37667187.81500002	35740254.48999998	5.39
	2023	1	38858279.47499999	37667187.81500002	3.16
	2023	2	38687900.88999999	38858279.47499999	-0.44
	2023	3	40549842.92499999	38687900.88999999	4.81
	2023	4	37348376.69000003	40549842.92499999	-7.9
	2023	5	39996687.55999999	37348376.69000003	7.09
	2023	6	42578718.07499997	39996687.55999999	6.46
	2023	7	39277143.405	42578718.07499997	-7.75
	2023	8	41947177.40999999	39277143.405	6.8
	2023	9	41109909.64500002	41947177.40999999	-2

Identifying 'Star Doctors' per Region:

-- Q11) Identify the "Star Doctor" of each region. A Star Doctor is defined as
 -- the one who generated the absolute highest total revenue in their specific
 -- region. Return the Region, Hospital Name, Doctor Name, and Total Revenue.

• with doctor_revenue as

```
(select h.region, h.hospitalname, d.firstname as doctor_name, sum(b.totalcost) as total_revenue
from doctors d join admissions a on d.doctorid = a.doctorid
join billing b on a.admissionid = b.admissionid
join hospitals h on d.hospitalid = h.hospitalid
group by h.region, h.hospitalname, d.doctorid, d.firstname),
ranked_doctors as
(select region, hospitalname, doctor_name, total_revenue,
dense_rank() over (partition by region order by total_revenue desc) as rnk
from doctor_revenue)
select region, hospitalname, doctor_name, total_revenue
from ranked_doctors
where rnk = 1;
```

	region	hospitalname	doctor_name	total_revenue
▶	Central	AIIMS Regional	Dr. Aarti	9749071.085000003
	East	Narayana Medical Center	Dr. Aryan	9757909.599999998
	North	Apex Health Centre	Dr. Amit	10417263.129999995
	Northeast	GNRC Hospitals	Dr. Karan	9123166.755000003
	South	Max Wellness Institute	Dr. Swati	9790747.409999998
	West	Sunrise Medical City	Dr. Manish	10166310.92

5. Dashboard in Power BI

An interactive, multi-page dashboard was built in Power BI to present insights visually to hospital administrators.







6. Strategic Recommendations

- **Resource Allocation:** Pediatrics and Orthopedics have the highest admission volumes; prioritize staffing and bed capacity in these areas.
- **Revenue Cycle Management:** A significant portion of bills are in 'Pending' or 'Defaulted' status. Implement stricter upfront insurance verification processes.
- **Regional Focus:** The South region generates the highest gross revenue. Consider expanding specialized services in southern network hospitals like KIMS General and Max Wellness Institute.
- **Supply Chain & Inventory:** AB+ and O- are the most prominent patient blood types; align the hospital blood bank procurement directly with this demographic breakdown to ensure readiness.